

Fig. 5: Circuit diagram of the RF remote

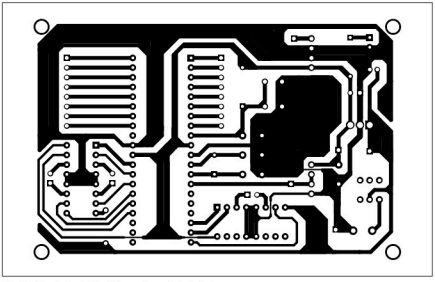


Fig. 6: Actual-size PCB of the main module (robot)

```
        printf("%c",i);
        }
For robot (receiver):
   while(1)
        {
        scanf("%c",&ii);
        _delay_ms(10);
        switch(ii)
        {
        case 'h':  //auto mode
        autonomous();
        break;
```

Software program

Programming of the AVR is done using embedded C language. It is similar to C language but includes all functionalities of C as well as access to AVR pins, peripherals and controls. C code is converted to hex code using WinAVR.

Hex codes generated are burnt into MCUs for the main module (robot) and the remote module. Working of the program is explained as comments in the main module (robot) and remote module source codes.

The program jumps to the main function where the object code actually starts. At remote module, a DIP (DIP1) switch is interfaced using which you can select operating mode (autonomous or manual), switching of the water pump, reset all settings or LED indicators test mode as given in Table I.

Tools used are described below:

WinAVR. WinAVR is a suite of executable, open source software development tools for Atmel AVR series. It includes GNU GCC compiler for C and C++, Programmer's Notepad, Makefile, etc.

Programmer's Notepad (PN). This is a source editor with some IDE features. PN can call any command-line tool and capture its output. It is ideal for calling make utility, which executes make file, which, in turn, calls the compiler, linker and other utilities used to build your software.

Makefile. WinAVR now includes MFile utility, an automatic make file generator for AVR GCC, which can run on various platforms including Windows, FreeBSD and Linux. You can use this utility to quickly generate make files for your project based on some simple menu input.

Construction and testing

An actual-size, single-side PCB of the main module is shown in Fig. 6 and its component layout in Fig. 7.

An actual-size, single-side PCB of the RF remote module is shown in Fig. 8 and its component layout in Fig. 9.

An actual-size, single-side PCB of the sensor module is shown in Fig. 10 and its component layout in Fig. 11.

You can also assemble the circuit on a general-purpose PCB. Before mounting the MCUs on PCBs, burn respective codes into MCUs using a suitable programmer board.

After mounting all components on PCBs, switch on the power supplies of respective units.

At the main module (robot), initially four LED indicators glow on even pins of PORTB of IC2 by default, and the robot waits for the signal from RF remote. You can also test all